



VLA-Corrector

**Lightweight Detect-and-Correct Inference
for Adaptive Action Horizon**

Embodied AI

Vision-Language-Action

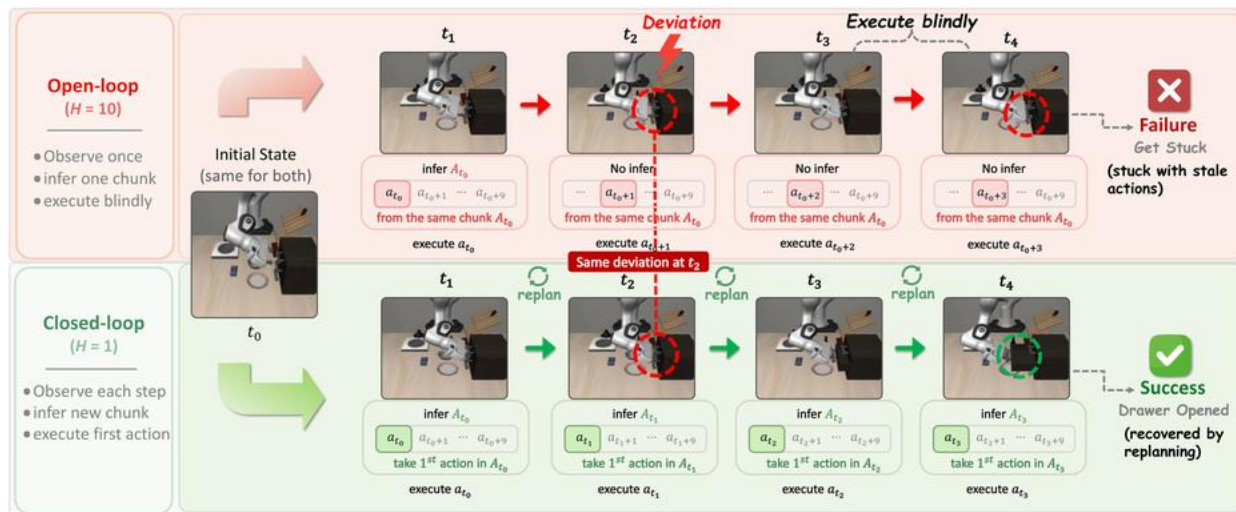
Action Correction

~40M Corrector

Motivation

Can a frozen VLA keep long-horizon efficiency while recovering from execution drift before stale actions accumulate?

Open-loop VLA execution leaves blind spots.



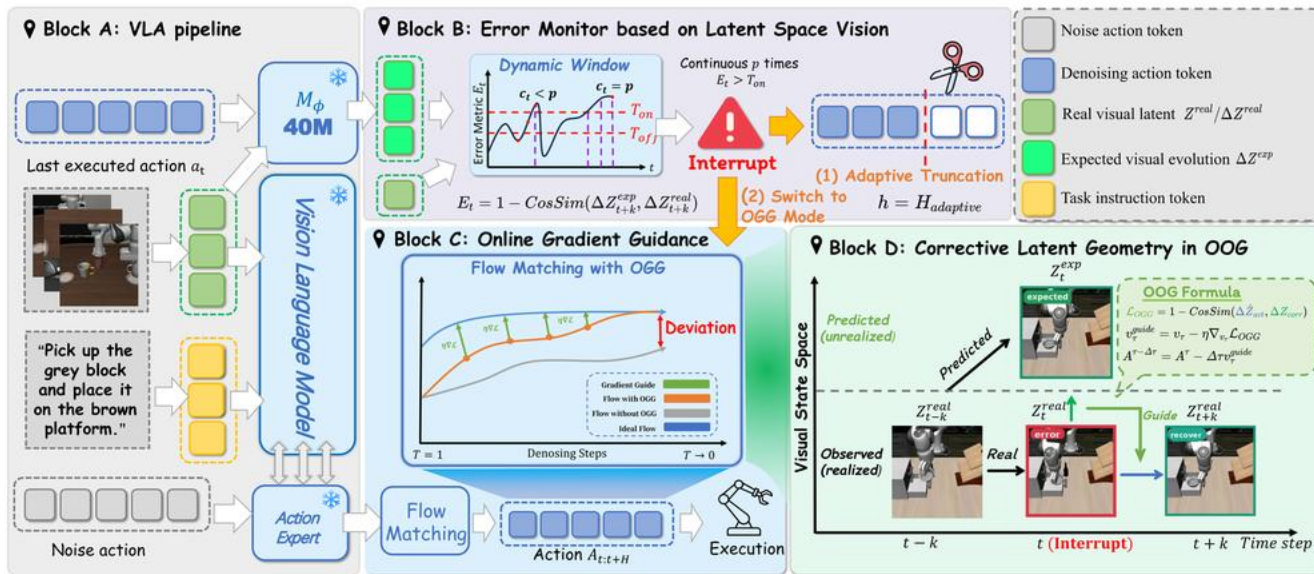
Long action chunks

Efficient VLA calls, but queued actions can become stale after drift.

Per-step replanning

More reactive, but expensive for large VLA policies.

Detect drift, truncate stale chunks, guide recovery.



01 Frozen VLA backbone with an external latent dynamics corrector.

02 Latent-space Vision Monitor detects persistent visual dynamics mismatch.

03 Online Gradient Guidance is applied only to the recovery query.

A small external model instead of full VLA retraining.

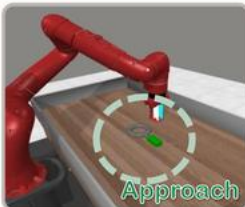
Critical phase

Precision-sensitive operations



Non-Critical phase

High-tolerance translational motions



Trainable component

~40M

Residual MLP corrector reported at roughly 38–42M parameters.

Policy backbone

Frozen

The base VLA is not fully retrained for correction.

Action horizon

Adaptive

Stable chunks continue; stale chunks are interrupted.

Reported gains across simulation and real robots.

MetaWorld / PI0.5

+15.65 pts

48.70% -> 64.35% avg.
success

LIBERO / PI0.5

+3.80 pts

94.00% -> 97.80% few-shot
success

AgileX PiPER

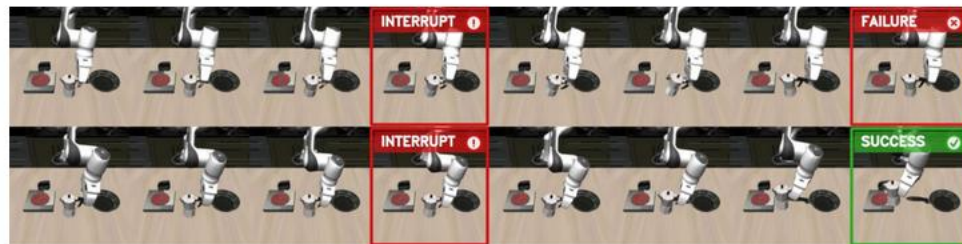
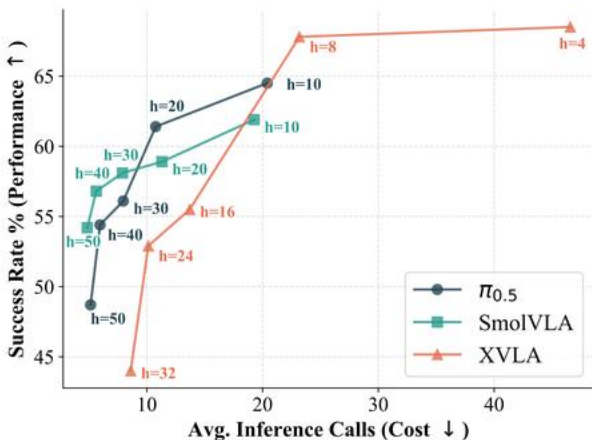
+17.7 pts

55.6% -> 73.3% real-world
success

Critical phases

83.7%

Truncations in
manually labeled
critical phases



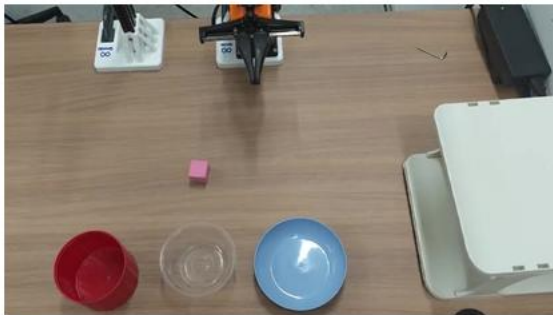
Qualitative recovery: truncate stale chunk, replan with OGG, complete the task.

Perturbation demos are kept as silent project-page clips.



Drawer alignment

Human moves the drawer during execution.



Block to blue bowl

Target bowl is shifted mid-task.



Block to white bowl

Object/target perturbation during execution.

The presentation uses static thumbnails; the project page still hosts the three silent demo videos.



VLA-Corrector

Resources

Code

<https://github.com/ZJU-OmniAI/vla-corrector>

Project page

<https://zju-omni.ai/github.io/vla-corrector/>

Paper

Coming soon

arXiv

Coming soon

Editable slides

[docs/assets/presentation/vla_corrector_presentation.pptx](#)